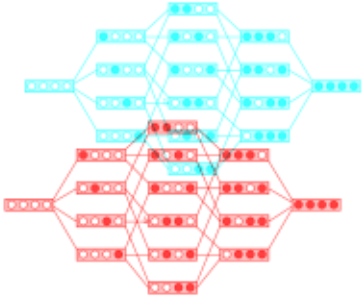
Introduction to Machine Learning

Feature Selection: Wrapper methods

Feature Selection: Wrapper methods



Learning goals

- Understand how wrapper methods work
- Understand how they can help in feature selection
- Know their advantages and disadvantages

Learning goals

- Understand how wrapper methods work
- Forward + backward search, EAs
- Advantages and disadvantages

OBJECTIVE FUNCTION

Given p features, **best-subset selection problem** is to find subset $S \subseteq \{1, \dots p\}$ optimizing objective $\Psi : \Omega \to \mathbb{R}$:

$$S^* \in \arg\min_{S \in \Omega} \{\Psi(S)\}$$

- Ω = search space of all feature subsets $S \subseteq \{1, \dots, p\}$. Usually we encode this by bit vectors, i.e., $\Omega = \{0, 1\}^p$ (1 = feat. selected)
- Objective Ψ can be different functions, e.g., AIC/BIC for LM or cross-validated performance of a learner
- Poses a discrete combinatorial optimization problem over search space of size = 2^p, i.e., grows exponentially in p (power set)
- Unfortunately can not be solved efficiently in general (NP hard; see, e.g., Natarajan 1995.)
- Can avoid searching entire space by employing efficient search strategies, traversing search space in a "smart" way

GREEDY BACKWARD SEARCH

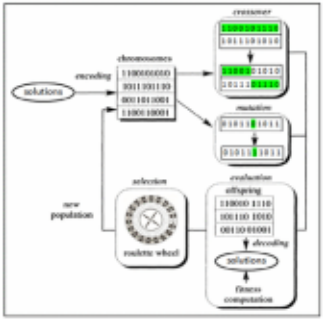
- Start with the full index set of features $S = \{1, \ldots, p\}$.
- For a given set S generate all $S_j = S \setminus \{j\}$ with $j \in S$.
- Evaluate the classifier on all S_j and use the best S_j.
- Iterate over this procedure.
- Terminate if:
  - the performance drops drastically, or
  - falls below given threshold.

- GFS is much faster and generates sparser feature selections
- GBS much more costly and slower, but sometimes slightly better.

EXTENSIONS: GENETIC ALGORITHMS FOR FS

Example Template for $(\mu + \lambda)$-Evolutionary Strategy applied to FS

1. Initialization: μ random bit vectors (feature inclusion/exclusion)
2. Evaluate model performance for bit vectors
3. Select μ fittest bit vectors (parents)
4. Generate λ offspring applying crossover and mutation
5. Select μ fittest bit vectors from $(\mu + \lambda)$ options for next generation
6. Repeat steps 2-5 until stopping criterion is met



Mitsuo Gen 1996

- Use CV/validation set for evaluation to avoid overfitting
- Choice of μ and λ allows some control over exploration vs. exploitation trade-off
- See our sids-lmu 2021 for further information